Journal of Comparative Economics 53 (2025) 118–138

Contents lists available at ScienceDirect

Journal of Comparative Economics

journal homepage: www.elsevier.com/locate/jce

Foreign-assisted infrastructure and local employment: Evidence from China's aid to Africa

Jiafu An [a], Shiqi Guo [b,*], Haicheng Jiang [c]

[a] Department of Real Estate and Construction, University of Hong Kong, Hong Kong SAR, China
[b] Antai College of Economics and Management, Shanghai Jiao Tong University, China
[c] School of International Trade and Economics, Anhui University of Finance and Economics, China

ARTICLE INFO

JEL Classification:
F35
J20
O18
O55
R23

Keywords:
Africa
Aid projects
China
Education
Employment
Foreign aid
Infrastructure
Labor
Productivity
Migration

ABSTRACT

This study examines the impact of Chinese infrastructure aid on local employment across ten African countries, surveying over 500,000 individuals from 2000 to 2014. Utilizing variations in proximity to aid projects and the timing of project construction relative to local surveys reveals that Chinese aid increases local employment by two percentage points in areas near project sites compared to those awaiting project commencement, contrasting with regions without such aid. Employment rises with the onset of construction, primarily benefiting individuals with lower educational attainment, and persists post-completion, sustained by the migration and employment of skilled workers. Various infrastructure projects temporarily boost local employment, with long-term benefits particularly pronounced for projects in education, healthcare, and water and power utilities. Cross-sectoral and spatial spillover effects are also identified, with results remaining robust after accounting for additional local development factors.

1. Introduction

Foreign aid is a crucial tool used by the international community to promote economic growth in developing countries. Despite trillions of dollars provided by Western donors, the effectiveness of such aid remains inconclusive (Easterly, 2003; Clemens et al., 2012; Qian, 2015; Galiani et al., 2017). Meanwhile, China has emerged as a significant player, committing over $350 billion in official finance to 140 countries between 2000 and 2014 (Dreher et al., 2021). The scale of Chinese aid has grown substantially, with The State

We extend our gratitude to the Editor, Timur Kuran, and the two anonymous referees for their insightful comments, which have greatly enhanced this paper. We are also thankful to Jean-Louis Arcand, Arcangelo Dimico, James Fenske, Matthias Flueckiger, Jianan Li, Pinghan Liang, Lin Lin, Ming Lu, Difei Ouyang, Albert Park, and Min Wang, along with participants at various conferences and seminars, for their valuable feedback and discussions. An earlier version of this paper was circulated as "Chinese aid and local employment in Africa." This work is supported by the National Natural Science Foundation of China (72203144, 72073098, 72273088). Jiafu An acknowledges financial support from the University of Hong Kong Start-up Fund for New Professoriate Staff, and Shiqi Guo acknowledges support from the Shanghai Pujiang Program.

* Corresponding author at: Antai College of Economics and Management, Shanghai Jiao Tong University, 1954 Huashan Road, Shanghai, China.
E-mail address: shiqiguo@sjtu.edu.cn (S. Guo).

https://doi.org/10.1016/j.jce.2024.11.003
Received 21 February 2024; Received in revised form 29 October 2024; Accepted 26 November 2024
Available online 5 December 2024
0147-5967/© 2024 Association for Comparative Economic Studies. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Council (2015, 2016) reporting that by the end of 2009, foreign aid totaled $41 billion, with an additional $15 bill\ion disbursed between 2010 and 2012.[1]

The rapid expansion and distinct approach of China's aid has fueled growing research interest. As one of the largest bilateral aid donors, and arguably the most controversial one, China emphasizes large-scale infrastructure in its official assistance, demonstrating notable efficiency in project design and completion. Characterized by a policy of non-intervention in the internal affairs of recipient countries, China's aid is largely demand-driven and free from explicit political conditions (The State Council, 2015). However, the existing literature mainly studies its impact on political capture and commercial entanglements (Dreher et al., 2019; Brautigam, 2011), with few exceptions that look at the macro-economic impact of Chinese aid (Xu et al., 2020; Dreher et al., 2021).

Chinese aid projects, particularly in Africa, have been linked to economic growth, as measured by GDP and nighttime light density (Xu et al., 2020; Dreher et al., 2021). Beyond macroeconomic impacts, studies have highlighted significant social effects, showing that China's foreign aid can reduce regional violence (Gehring et al., 2022), enhance social welfare (Martorano et al., 2020), influence ideological alignment among recipient citizens (Blair et al., 2022), affect union involvement (Isaksson and Kotsadam, 2018b), and potentially interact with local norms to increase corruption (Isaksson and Kotsadam, 2018a).

Despite these findings, the mechanisms driving these individual-level impacts remain unclear. To address this gap, our paper examines how individuals in recipient countries engage with Chinese projects, focusing on the effect of Chinese infrastructure aid on local employment in Africa. Employment in sectors related to these projects directly exposes locals to Chinese aid, making it crucial to understand how such aid influences individual employment. This understanding is key to explaining the broader effects on the ideologies and norms of local populations and offers insights into the aggregated impact on economic growth.

Moreover, our study contributes to the broader understanding of the labor market impact of infrastructure construction in Sub-Saharan Africa (SSA), a region that significantly lags behind other developing areas, trailing the International Development Association (IDA) average by at least 20 percentage points on key infrastructure indicators (Foster et al., 2009). With an estimated annual infrastructure funding gap of $22 billion (Go and Page, 2008), African leaders have sought international support to address this shortfall. While recent studies have examined the impact of specific infrastructure projects in individual countries (Dinkelman, 2011; Fiorini and Sanfilippo, 2022; Ratledge et al., 2022; Perra et al., 2024), there is limited understanding of how these findings apply across different SSA countries with diverse socioeconomic contexts. Comparative analyses of various infrastructure types within a consistent economic and political framework are also scarce. Chinese-funded infrastructure projects, which span multiple countries and include various types of infrastructure, provide a unique opportunity to fill these gaps. Additionally, by analyzing the timing of both project initiation and completion, we can distinguish between the effects during construction and those following infrastructure improvements.

Identifying the causal impact of Chinese aid on employment presents a challenge due to the non-random placement of aid projects. Dreher et al. (2019) demonstrate that the allocation of Chinese aid can be influenced by the political interests of leaders in recipient countries. Furthermore, some argue that the placement of Chinese aid is driven by China's economic interests, particularly in securing natural resources in Africa (e.g., Alden, 2005; Naim, 2009). Cervellati et al. (2022) further reveal that the location of Chinese projects may be affected by geo-climatological conditions, as these factors impact where Chinese personnel choose to settle. If such factors are associated with local employment, then a simple comparison of labor outcomes in areas with and without Chinese aid projects could suffer from omitted variable bias. Moreover, reverse causality is a concern if areas with high labor availability tend to attract Chinese-funded infrastructure projects.

To address empirical challenges, we utilize the coincidental timing of Demographic and Health Survey (DHS) fieldwork and the implementation phases of Chinese infrastructure aid projects. We combine geocoded data from over 500,000 individuals surveyed by the DHS between 2000 and 2014 with information on 360 Chinese infrastructure projects classified as Official Development Assistance (ODA-like) from AidData. This integration enables us to determine whether individuals live in proximity to Chinese infrastructure projects, which potentially affect their livelihoods. We then examine how these projects influence local employment over time by analyzing the periods between the project initiation and nearby survey fieldwork.

Our approach, grounded in recent aid literature, employs a difference-in-differences framework to compare individual employment outcomes in regions with ongoing or completed Chinese aid projects against those in regions soon to receive aid, as well as regions without such projects. This method enables us to scrutinize potential biases in Chinese aid placement concerning employment outcomes by contrasting soon-to-be treated areas with those that never receive treatment. Although this approach is common in recent studies examining the local effects of aid and mineral projects,[2] its nuances, identifying assumptions, and methods for addressing additional identification threats have not been thoroughly explored in the scholarly discourse.

We demonstrate that by controlling for fixed effects specific to each country-by-year and subnational region, which address selection bias due to local time-invariant factors and characteristics that change over time at the country level, the coincidental timing of surveys and aid projects generates plausibly exogenous variations in local exposure to Chinese aid. To verify this premise using repeated cross-sectional survey data, we establish that individuals in treated locations have comparable characteristics to those in soon-to-be-treated locations, conditional on the fixed effects mentioned above. We further corroborate that the timing of DHS surveys

[1] Between 2000 and 2014, U.S financial aid to Africa (the largest bilateral donor) rose from $2.11 billion to $9.33 billion. During the same timeframe, Chinese aid increased from $0.67 billion to $6.19 billion. These figures are derived by the authors using the World Development Indicators and data from AidData.

[2] See, for example, Isaksson and Kotsadam (2018a, 2018b), Knutsen and Kotsadam (2020), Isaksson (2020), and Cruzatti, Dreher, and Matzat (2023).

relative to the start of aid projects, which determines residents' treatment status, is exogenous and not influenced by local time-varying factors. Evidence also indicates parallel trends in employment outcomes between treatment and control areas before project initiation.

Our paper presents a comprehensive analysis of the nuanced impacts of Chinese aid on local employment, revealing several key findings. First, we observe a 1.5 to 3.1 percentage point increase in employment within 50 km of aid projects during the first two years, with an additional 1.3 percentage point increase after six years. These effects translate to a 3.5 % overall rise in employment, both during the construction phase and following infrastructure improvements, compared to the sample mean. Importantly, the construction phase primarily benefits less educated workers, while the post-construction phase favors those with higher education levels.

The positive net effect on employment spans multiple dimensions. Across occupations, we find a significant increase in skilled manual labor and service sector employment, alongside a decline in agricultural employment. Over time, there is a consistent shift favoring skilled labor, indicating a broader transition from unskilled to skilled employment. Additionally, regions hosting Chinese aid exhibit a clear shift towards formal employment, characterized by stable, year-round jobs with monetary wages and training opportunities, as opposed to the sporadic and often non-monetary nature of informal employment. These trends suggest that large-scale infrastructure development may be driving a structural transformation in local labor markets.

Moreover, the employment impact of Chinese aid diminishes with distance, disappearing beyond 200 km, indicating that employment growth is localized rather than causing displacement in nearby areas. After project completion, there is a notable migration of skilled and highly educated workers towards project areas, suggesting that Chinese aid may activate the potential of previously underutilized labor forces in other areas. Thus, in addition to the structural transition underlying the long-term employment increase, the immigration of skilled labor into local areas presents another mechanism driving sustained employment growth.

The impact of Chinese aid also varies by infrastructure type. While transportation projects yield a greater short-term employment boost, projects related to water, power, hospitals, and schools have a more pronounced effect on local employment over a longer period. Our results remain consistent even when accounting for potential influences from other official Chinese finance, infrastructure aid from other donors, local investment access, political or ethnic favoritism, and various empirical specifications.

This paper contributes to several strands of literature by enhancing our understanding of the effects of infrastructure investment on local labor market in Africa. Previous studies have shown that domestic infrastructure initiatives, such as roads, electrification, and telecommunications projects, significantly boost employment across the continent. For example, Dinkelman (2011) found that electrification in South Africa increased female employment by reducing the time spent on household chores, while Hjort and Poulsen (2019) discovered that faster internet selectively improves employment prospects for skilled workers without affecting unskilled labor. Additionally, Perra et al. (2024) found that road improvements in Ethiopia led to better connectivity and increased competition for local firms, though this did not uniformly benefit the economy. While the formal sector is associated with higher labor productivity, the informal sector did not experience a similar increase.

Our research builds upon this foundation in two significant ways. First, we utilize a unique dataset detailing the start and completion times of Chinese aid projects, enabling us to examine both the immediate employment boost and the longer-term benefits that follow infrastructure improvements. While there is a perception that Chinese projects do not substantially enhance local labor markets due to reliance on expatriate workers (Zhao, 2014; Wegenast et al., 2019), recent surveys challenge this view.[3] However, causal evidence on how local populations benefit from large-scale infrastructure construction financed by China's development aid remains scarce. Our findings confirm a short-term increase in manual labor employment in areas close to these projects, as well as significant positive spillover effects on service sectors and neighboring areas during construction. Furthermore, we provide insights into the mechanisms behind long-term employment growth, revealing trends in which skilled and highly educated workers migrate to areas post-improvement, and educational attainment of young individuals rises, signaling a transition from unskilled to skilled labor alongside these aid projects.

Second, we provide a comparative analysis of the effects cross multiple types of infrastructure. While prior literature has explored the employment impacts of different infrastructure within specific exogenous contexts, typically focusing on single countries or specific infrastructure types, the shared socio-political framework of China's aid enables a unique comparison analysis across various infrastructure types. Our analysis indicates that while all infrastructure investments generally yield positive employment effects, the magnitude of these effects varies by type. Specifically, transportation projects tend to provide a greater immediate employment increase, whereas investments in water and power facilities, hospitals, and schools have more substantial long-term employment benefits, which may contribute to sustained development.

Furthermore, our paper presents a nuanced perspective of Chinese aid's impact on recipient countries. While economic theory suggests that capital inflows and infrastructure investments—often forms taken by Chinese aid in Africa—can foster growth (e.g., Domar, 1946; Romer, 1986; Mankiw et al., 1992; Delong and Summers, 1992), much of the current literature has focused on the negative, unintended social consequences. Studies have reported adverse effects on local governance (Dreher et al., 2019; Anaxagorou et al., 2020; Gehring et al., 2022; Ping et al., 2022), harmful social norms (Isaksson and Kotsadam, 2018a, 2018b; Isaksson, 2020), and impediments to economic reforms (Brazys and Vadlamannati, 2021). Dreher et al. (2021) stand out by showing that Chinese aid promotes short-term country-level economic growth through public capital formation. However, there remains a gap in our understanding of the microfoundations and dynamics behind the economic impact of China's development aid. By combining aid projects with household data from extensive cross-country surveys, we elucidate how local employment is affected—a key factor in economic

[3] For instance, Sautman and Yan (2015) found that 85 % of the workforce in Chinese firms operating in Africa consists of local Africans. Similarly, a McKinsey & Company study involving 1,073 Chinese companies across Africa revealed that local hire represented 89 % of the workforce, amounting to approximately 300,000 positions (Sun et al., 2017).

growth models linking capital investment to productivity increases (e.g., Solow, 1957; Romer, 1986). Beyond mapping employment dynamics, we also explore potential mechanisms behind longer-term employment benefits and demonstrate how these gains spill over into other sectors and regions. These insights contribute to a more comprehensive understanding of China aid's impact on recipient countries.

The remainder of this paper is organized as follows: Section 2 introduces the data and outlines the identification strategy. Section 3 discusses the baseline results, including the impacts on individual employment status, as well as dynamic and spatial spillover effects. Section 4 presents the heterogeneous effects by infrastructure type, education level, and labor skills. Section 5 examines the potential influence of other development resources and the broader impact of Chinese aid on household wealth and nightlight emissions at the local level. Section 6 concludes the paper.

2. Data and identification

2.1. Data

We utilize two main data sources to assess the impact of Chinese aid on individual employment in Africa. The data on Chinese aid projects are sourced from AidData's Geocoded Global Chinese Official Finance dataset (version 1.1.1), which covers the years 2000–14 (AidData Research and Evaluation Unit, 2017). This comprehensive dataset aggregates information from public records, providing detailed accounts of Chinese aid initiatives.[4] Our main analysis focuses on projects intended for development, classified as Official Development Assistance (ODA-like) in alignment with existing literature on Chinese aid.[5] These projects encompass infrastructure development, including transportation networks, educational and healthcare facilities, and water and power infrastructure, among other public and social services.[6] We leverage data on the geographic locations and initiation years of these projects.[7]

Employment data are derived from the DHS in Africa, which offer repeated cross-sectional information on various household characteristics.[8] These surveys provide detailed information on household members' employment status, such as current work activity, job type, duration of employment, and payment methods. Given the scarcity of detailed official employment data in Africa, the multiple waves of DHS surveys allow for an in-depth examination of employment at a disaggregated level.

We combine these datasets based on (i) the proximity of aid projects to DHS clusters and (ii) the timing of DHS fieldwork relative to the commencement of nearby aid projects. By assessing the distance from each DHS cluster to Chinese aid initiatives, we determine whether a cluster is close to, and potentially affected by, nearby projects. We then assess how the influence of Chinese aid on local employment evolves over time by analyzing the periods between the project start years and the survey years across various DHS waves. To accurately gauge this dynamic effect, we observe employment across multiple periods in the repeated cross-sectional data before and after the initiation of nearby Chinese aid projects.[9] Consequently, our analysis focuses on the top ten African countries that received the majority of Chinese aid projects and hosted at least two rounds of geocoded DHS surveys during the data period.[10]

Our sample includes 360 Chinese ODA-like infrastructure aid projects from 2000 to 2014, alongside 29 DHS surveys conducted around this period (1999–2016). These projects represent 54.63 % of all Chinese infrastructure aid in Africa. Among them, there are 103 road and railway projects, 69 water and power facility projects, 60 schools and hospitals, and 128 other types of infrastructure projects. The corresponding DHS surveys include data on 559,724 individuals from 312,460 households within 15,101 clusters in ten SSA countries (Cameroon, DR Congo, Ethiopia, Ghana, Kenya, Namibia, Nigeria, Tanzania, Uganda, and Zambia). These nations accounted for approximately 57 % of SSA's population and 46 % of its GDP as of 2012. For further details on the aid projects and DHS

[4] See Strange et al. (2014) for a detailed explanation of the data collection methodology.

[5] While China does not explicitly distinguish between Official Development Assistance (ODA) and Other Official Flows (OOF), AidData categorizes 76 % of Chinese-funded projects in Africa as ODA-like, 12 % as OOF-like, and 12 % as Vague Flows (VF). Our baseline analyses focus on ODA-like projects because they are concessional and align with the traditional concept of Western foreign aid. This allows us for a direct comparison with the extensive literature on the impacts of traditional foreign aid and the recent studies on Chinese aid. OOF projects, which typically involve larger financial commitments, are less concessional and more commercially oriented, with less uniform financing and implementation processes compared to ODA projects. In Section 5.1, we extend our analysis to include other categories of Chinese aid.

[6] Among the various Chinese aid initiatives, we identified 225 projects focused on grants, food and equipment donations (46 %), medical teams and volunteer services (46 %), technical guidance (6 %), and scholarships for African students and debt exemptions (2 %). Since these projects do not involve physical infrastructure construction, we have excluded them from our main analyses. The different categories of aid projects are presented in Table A1 in the Appendix, while Table A2 shows the number of projects by country.

[7] To ensure the precision of our data regarding the initiation time and location of aid projects, our study is limited to those with a specific start year and a location precision code of 1 or 2 in the AidData, indicating a maximum deviation of 25 km of an exact location.

[8] We utilize standard DHS surveys with geocoded information on survey clusters, enabling precise measurement of household proximity to Chinese aid projects. However, this geocoded information is unavailable for 253 DHS clusters (1.7 %). We exclude them from our analysis.

[9] For example, if a country received a single Chinese aid project in 2005 and DHS surveys were conducted in both 2005 and 2010, our analysis would be limited to assessing the local impact of the aid project for that year and five years subsequent to its commencement.

[10] Expanding our dataset to include additional African countries is not expected to significantly alter our findings. This is primarily because, beyond the top ten recipient countries in our sample, other nations received fewer Chinese aid projects and would mostly constitute the control group in our empirical framework due to this scarcity of projects. The robustness checks presented in Table B1 affirm that our findings remain consistent even when we exclude the control group and exploit the staggered deployment of Chinese aid, relying solely on areas close to Chinese aid for identification. Consequently, the inclusion of more control group observations from the other African countries is unlikely to affect our findings.

waves included in our sample, please refer to Appendix A.[11] Fig. 1 illustrates the geographical distribution of Chinese aid projects in relation to DHS clusters.

2.2. Identification strategy

After combing DHS survey data with Chinese aid projects based on their locations and relative timing, we can categorize the matched DHS clusters into three distinct groups. The first group, designated as the control group, contains 9236 clusters that did not observe any Chinese aid projects within a 50 km radius throughout our sampling period. Consequently, it is unlikely that the local residents in these clusters were influenced by Chinese aid.[12] The second group, termed the post-treatment group, comprises 4331 DHS clusters that were within a 50 km radius of one or more aid projects initiated *before* the DHS survey was conducted. Since the presence of Chinese aid projects is the treatment of interest, individuals in these clusters had experienced the implementation of Chinese aid projects and thus had been treated at the time of the survey. The third group, known as the pre-treatment group, includes 1534 clusters where aid projects within 50 km were to take place in the future but had not commenced at the time of the survey. The DHS surveys for this group reflect the conditions of the local residents before any treatment effect.

To identify the impact of Chinese aid projects on local employment, we use the following regression specification:

$$Y_{ist} = \beta_0 Pre_{st} + \beta_1 Post_{st} + X_i\gamma + u_z + \nu_{ct} + w_{ist} \tag{1}$$

The dependent variable Y_{ist} indicates whether individual i from cluster s was working at the time of the survey in year t. The dummy variable $Post_{st}$ represents the post-treatment group; it equals one if cluster s, surveyed in year t, falls within 50 km of any Chinese infrastructure aid projects that had started before year t. Pre_{st} is the dummy variable for the pre-treatment group; it equals one if cluster s is within 50 km of aid projects that started only in or after year t.[13] In other words, for the pre-treatment group, no projects within 50 km had occurred before the survey was conducted. The control group—clusters without proximate Chinese aid projects during the sample period—is omitted from the regression and serves as the benchmark for comparison.

The covariate vector X_i encompasses individual-level factors (such as gender, age, ethnicity, and years of education)[14] that could influence employment outcomes, as well as cluster-level characteristics including rural/urban status, terrain roughness, travel time to the nearest city, distance to capitals, national borders, water bodies, and protected areas, ethnic diversity, presence of natural resources like gold, copper, diamond, and petroleum, along with predetermined socio-economic indicators such as conflict intensity, population density, and night light density.[15] These covariates are intended to capture the local development potential and conditions that could affect the distribution of Chinese aid.

Fixed effects are denoted by u_z for region-specific (second-level subnational administration, or ADM2 level) variations and ν_{ct} for country-year variations.[16] These control for time-invariant regional attributes (e.g., culture, social, and economic needs, connection

[11] In Appendix A, Tables A1 and A2 detail the distribution of Chinese aid projects across different infrastructure categories and recipient countries, respectively. Table A3 outlines the DHS waves utilized in our paper. Fig. A1 illustrates the distribution of observations within the pre- and post-treatment groups across different countries. The sample is generally balanced concerning the number of observations in each group for each country, with notable exception of DR Congo, where there is a notably smaller number of pre-treatment observations. In our robustness checks in Table B1, we exclude the 7.1 % of observations from DR Congo to demonstrate that the results are not biased by this imbalance. Fig. A2 displays the count of observations linked to different types of infrastructure within the post-treatment group across various countries. With an exception of Nigeria, where only two categories of Chinese infrastructure aid were observed, every other country in the study witnessed at least three types during the sampling period. For robustness, we drop Nigeria's data in Table C2 to assess the heterogeneous impact of infrastructure on employment.

[12] A 50 km radius has been selected as the cutoff in our main analysis as it represents a practical commuting distance within African countries (Knutsen et al., 2017). In Section 3.3, we adjust this radius incrementally to evaluate the spatial spillover effects of Chinese aid projects.

[13] When Chinese aid coincides with DHS surveys within the same year, it is unclear whether the aid started before or after the surveys due to the missing exact start dates of most projects. For a conservative approach, we categorize such instances (1.9 % of our observations) as pre-treatment. The robustness of results is unaffected by the exclusion of these cases.

[14] Regarding ethnicities, we incorporate 433 dummy variables for different ethnic groups. referencing the digitized map by Nunn (2008) based on Murdock's (1959) classification, which accessible at https://scholar.harvard.edu/nunn/pages/data-0.

[15] The DHS dataset provides key geographical characteristics for the clusters. We calculate distances from each cluster to the national capital and quantify ethnic diversity within 50 km using Murdock's (1959) classification. For natural resources, we include the number of gold, copper, and diamond deposits within 50 km, using data from the US Geological Survey's Major Mineral Deposits of the World (https://mrdata.usgs.gov/major-deposits/). The presence of petroleum resource within 50 km is indicated by a dummy variable from the Petroleum Dataset v.1.2 (1946-2003; https://www.prio.org/Data/Geographical-and-Resource-Datasets/Petroleum-Dataset/). Conflict events are sourced from the UCDP Georeferenced Event Dataset Global version 19.1 (https://ucdp.uu.se/downloads/index.html#ged_global), defined as any incident with at least one direct death caused by organized actors. We analyze conflict events from1989 to 1999 within 50 km from DHS clusters. Finally, we include population and nighttime light density covariates, using their logarithmic values from the DHS dataset for each cluster in 2000.

[16] Our study encompasses 1,955 second-level subnational administrative regions, classified as ADM2 by the Global Administrative Areas database (available at https://gadm.org/maps.html).

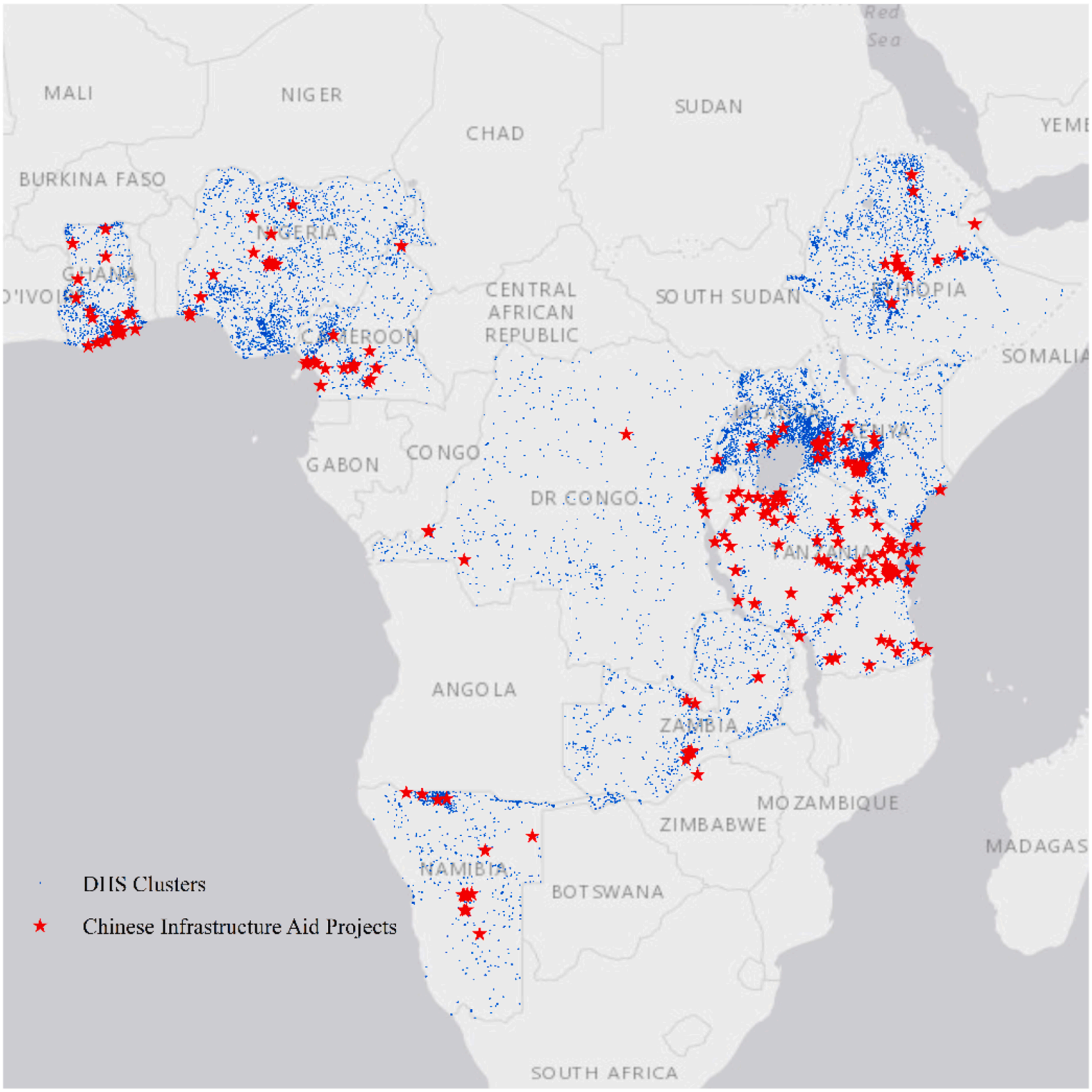


Fig. 1. Geographical distribution of Chinese aid projects and DHS clusters.
Notes: This figure shows the distribution of Chinese Official Development Assistance (ODA-like) infrastructure aid projects from 2000 to 2014, alongside DHS survey clusters in the ten African countries included in our sample: Cameroon, DR Congo, Ethiopia, Ghana, Kenya, Namibia, Nigeria, Tanzania, Uganda, and Zambia.

with political leaders) and dynamic factors such as bilateral relations with China and recipient countries' political and economic changes over time. These two sets of fixed effects largely control for factors that could potentially influence the distribution of Chinese aid, such as diplomatic relations between recipient countries and China, political affiliations with national leaders, and local economic demands (e.g., Dreher et al., 2018, 2019; Guillon and Mathonnat, 2020).[17] Standard errors are clustered at the level of DHS clusters, the primary sampling units of DHS.

The difference between β_1 and β_0 yields a difference-in-differences (DiD) type of estimate of Chinese aid's impact on employment. Specifically, β_1 reflects the employment status of post-treatment clusters relative to the control clusters, while β_0 represents the pre-treatment employment in comparison to the control group. With the fixed effects described above, these comparisons—both post-treatment versus control and pre-treatment versus control—are conducted within each ADM2 region and country-by-year cell, as

[17] We also find evidence supporting the prevalence of ethnic patronage and political favoritism in SSA, as illustrated by infrastructure allocation. In regions sharing the ethnicity of national leaders, 62 % of the local residents resides within 50 km from a Chinese infrastructure aid project (wither existing or planned), whereas in other regions, this figure drops to 33 %. Furthermore, 47 % of residents in the birth regions of these leaders live within 50 km of Chinese aid projects, compared to 35 % in other regions. These findings align with existing literature on ethnic patronage and political favoritism in SSA countries (Burgess et al., 2015; Dreher et al., 2019).

illustrated in Appendix Figure A3. Thus, β_1 is estimated by comparing the post-treatment group with the equivalent control group within the same ADM2 region and country-year context. This comparison captures the difference between the treatment and control groups *after* the treatment. Similarly, β_0 is estimated by comparing the pre-treatment group to its control counterpart, reflecting the pre-treatment differential. The DiD estimator, $\beta_1 - \beta_0$, contrasts the pre-treatment difference with the post-treatment difference, thereby isolating the causal effect of Chinese aid projects on local employment.[18] Notably, a non-significant β_0 would suggest that site selection for Chinese aid does not corrlate with employment outcomes, incidating exogeneity in project placement.

Eq. (1) assesses the overall impact of aid projects on employment by comparing the groups based on the status of nearby aid projects at survey time.[19] To delve into the temporal dynamics of this impact, we employ an additional specification:

$$Y_{is,t} = \alpha_0 Pre_{s,t} + \sum_{\tau=1}^{T} \alpha_\tau Post_{s,t-\tau} + \alpha_{T+1} Post_{s,t-(T+1)} + X_i\gamma$$

$$+u_s + v_{ct} + w_{ist} \tag{2}$$

Distributed lagged terms such as $Post_{s,t-\tau}$ indicate exposure to aid projects initiated one year prior to survey year t, extending up to T years before. For example, $Post_{s,t-1}$ equals 1 if aid projects within 50 km of cluster s started in year *t-1*, indicating that individuals in these DHS clusters were exposed to Chinese aid projects for one year. Similarly, $Post_{s,t-2}$ equals 1 if aid projects within 50 km began two years before the survey. Coefficient α_0 measures the difference in employment outcomes between areas without aid projects and those where projects were scheduled but had not yet started. The differences $\alpha_1 - \alpha_0$, $\alpha_2 - \alpha_0$, and $\alpha_T - \alpha_0$ provide DiD estimates of the evolving impact on employment following project commencement. The term $Post_{s,t-(T+1)}$ captures the cumulative impact of projects initiated more than $T + 1$ years before, with $\alpha_{T+1} - \alpha_0$ measuring the long-term *aggregated* impact of aid projects after $T + 1$ years. The covariates and fixed effects remain consistent with Eq. (1).

Descriptive statistics are presented in Table A4. On average, 65 % of survey respondents were employed, with a mean age of 32 and an average of six years of education.[20] Approximately 32 % of respondents were male, and 63 % resided in rural areas.[21] Within the 50 km radius of aid projects, 27 % of individuals were exposed to these projects at the time of the DHS survey, while 9 % were surveyed before project initiation. For the remaining 64 %, no aid projects were present within a 50 km radius during our sample period.

2.3. Threats to identification

Our identification strategy faces a primary challenge due to the nature of the repeated cross-sectional DHS data, where each individual is observed only once. This contrasts with a standard DiD approach in panel data settings, where individuals are observed multiple times before and after treatment. In our context, individuals residing within 50 km of Chinese aid are categorized into either the pre-treatment or post-treatment group, based on the timing of project implementation relative to the DHS survey fieldwork; no individual is observed both before and after the initiation of local Chinese aid projects. Therefore, we must assume that there is no endogenous selection of individuals into pre- and post-treatment groups, implying that individuals' treatment status (i.e., pre- or post-treatment) should be determined exogenously.

The assignment to treatment status depends on the relative timing of the DHS survey and Chinese aid—if Chinese aid starts before the DHS survey, local residents fall into the post-treatment group; if the survey occurs first, they fall into the pre-treatment group. This assumption implies that the scheduling of the DHS survey and Chinese aid projects is independent. This type of identification strategy, relying on the independent timing of two events, has been employed in notable studies to draw causal inference. For instance,

[18] To verify our main identification approach, we conduct a conventional difference-in-differences analysis using aggregated panel data. We divide Africa into 0.5 × 0.5 degree grid cells and compile annual data on individuals and aid projects within these cells. This creates a grid-cell panel dataset that tracks employment changes within the same cell following the arrival of Chinese aid, compared to cells without such aid. The results, available upon request, align with our main approach, confirming the parallel trends assumption and are further validated by an additional instrumental variable strategy addressing potential selection bias.

[19] The specification of Eq. (1) aligns with the approach taken by Knutsen et al. (2017) and Isaksson and Kotsadam (2018a, b), which hinges on whether a local project is in an *active* phase.

[20] We calculate respondents' ages using their reported year of birth (age = year + 1 – birthyear) to account for different age calculation methods across societies. This method results in slight discrepancies with respondents' self-reported ages (correlation coefficient = 0.953). However, the results remain consistent when we use self-reported ages as a control.

[21] Women are overrepresented in the DHS data, as male questionnaires are often administered to a subsample of selected households. In our study, we pooled the female (IR) and male (MR) datasets. As shown in Table A4 in the Appendix, on average, one male in every three selected households was interviewed. Consequently, this overrepresentation of females may cause our results to reflect more on the effects of China's development aid projects on women's circumstances. For a robustness check, we employed adjusted sampling weights to correct for gender representativeness. DHS household weights vary across clusters, accounting for selection probability and response rates. In our combined household member (PR) datasets, 475,250 women and 419,174 men were eligible by age, but only 400,157 women and 202,200 men were ultimately subsampled. To correct for gender representativeness, we multiplied male respondents' sampling weights by 2.058932, following DHS guidance (source: https://userforum.dhsprogram.com/index.php?t=msg&goto=13158&S=Google). The results of this check are reported in Table B1 in the Appendix, showing that our original findings persist with only slightly reduced magnitudes. This indicates that the effects on female employment are significant. In an additional subsample analysis, we observe slightly greater effects on female employment, aligning with Benshaul-Tolonen's (2024) finding that even male-biased industries like gold mining improve female employment in mining communities.

Depetris-Chauvin et al. (2020) leveraged the exogenous timing of Afrobarometer surveys relative to national football matches to assess the impact of wins on national identity. Guo and An (2022) similarly utilized the timing of terrorist attacks in relation to Afrobarometer surveys to investigate their effect on future outlooks. In our case, it is highly unlikely that DHS teams coordinate with Chinese aid project teams, making their timing—and thus the treatment assignment—plausibly exogenous. We empirically test this assumption below.

Firstly, if treatment status is indeed exogenously determined—meaning selection into pre- and post-treatment groups is effectively random—then individuals in these two groups should be comparable in the absence of treatment. We test this by empirically examining (i) whether individuals in both groups share similar demographic characteristics and (ii) whether survey clusters assigned to treatment and control groups differ along various predetermined dimensions as of the base year 2000. The findings are presented in Table 1.

In Table 1, we start by unconditionally comparing individuals in aided DHS clusters (including both pre- and post-treatment groups) to those in unaided clusters (the control group) in column 1. Column 2 provides a conditional comparison, controlling for ADM2 and country-by-year fixed effects. Both columns indicate that Chinese aid projects are generally allocated to areas with younger, more educated populations, fewer males, closer proximity to urban areas, smoother terrain, higher population density, and greater nighttime light emissions. These findings suggest that a direct cross-sectional comparison between areas with and without aid projects may not yield an accurate estimate of the local impact of Chinese aid due to selection biases.

To further assess whether the variation we exploit for identification is plausibly exogenous, we compare pre- and post-treatment groups in columns 3 and 4. In column 4, we control for country-by-year and ADM2 fixed effects, thus comparing individuals within the same ADM2 region and country-by-year cell. We observe no significant statistical differences across all relevant demographic, geographical, and predetermined characteristics between pre- and post-treatment groups. This supports our contention that the relative timing of aid projects and DHS surveys provides a quasi-random exposure to treatment.

Secondly, we explore whether this relative timing is influenced by time-varying characteristics. Following Akerman et al. (2015), we examine this relationship using Eq. (3), which accounts for time-varying effects:

$$Timing_{st} = X_s\gamma + (\theta_t \times X_s)\delta_t + u_s + v_{ct} + w_{st} \tag{3}$$

Here, $Timing_{st}$ denotes the relative timing of the DHS survey to local Chinese aid initiation, calculated as the year of DHS survey minus the year when the nearest Chinese aid project commenced. X_s includes variables from Table 1 that represent demographic, geographical, and development indicators. θ_t represents year indicators; by interacting θ_t with X_s, we allow local factors to dynamically influence the relative timing between DHS surveys and proximate aid projects. Fixed effects for each ADM2 region (u_s) and country-by-year cell (v_{ct}) are also included. Appendix Figure A4 displays estimated coefficients δ_t over time for each local characteristic. As indicated in Figure A4, there are no consistent time-varying correlations between these important cluster-level characteristics and the relative timing, with only three out of 168 coefficients showing statistical significance. This suggests the absence of a systematic pattern where local factors might affect the relative timing of Chinese aid and DHS survey implementation in our sample.

Taken together, our results indicate that the timing between DHS surveys and Chinese aid projects is exogenous and unaffected by key cluster characteristics. This allows us to confidently use a DiD estimation approach with repeated cross-sectional survey data. Beyond this assumption, we rigorously test the parallel trend assumption in Section 3.

3. Main results

3.1. Baseline results

3.1.1. Graphical evidence

Our identification strategy relies on the geographic proximity of DHS clusters to Chinese infrastructure aid projects and the timing aid project implementation relative to DHS surveys. Fig. 2 displays the variation in employment rates based on the distance from the nearest Chinese project. For clarity, DHS clusters are categorized into 20 equal-sized bins. The local average for each bin is plotted as a single point—red for post-treatment and blue for pre-treatment groups—with corresponding quadratic fits for each group.

We observe that proximity to implemented Chinese aid projects (post-treatment) correlates with higher employment rates, while no such correlation exists in areas awaiting project implementation (pre-treatment). A noticeable difference in employment between pre- and post-treatment groups emerges only in proximity to Chinese aid projects, specifically within 150 km. Beyond this distance, this difference diminishes, equalizing the employment rates of the two groups. These patterns indicate a positive, localized impact of Chinese aid on employment.

Fig. 3 presents cluster-level employment rates against the timing of DHS surveys relative to the implementation of Chinese aid projects. This is depicted separately for clusters within 150 km of the project (red line) and those situated between 150 and 400 km away (blue line).[22] Prior to the initiation of Chinese aid projects, employment rates in proximal and distant areas were comparable and mostly parallel over time. However, post-implementation, areas within 150 km consistently experienced higher employment rates for several years. In the longer term (e.g., after eight years), these areas also tend to maintain higher employment levels. Overall, Fig. 3 indicates a positive influence of Chinese aid on local employment in both short- and medium-term perspectives.

[22] We select a 150 km radius as the benchmark for our analysis, as illustrated Fig. 2, where it is evident that the employment differences between pre- and post-treatment groups are pronounced at this range.

Table 1
Balance checks.

Differences	Aided clusters –Non-aided clusters		Post clusters –Pre clusters	
	Unconditional (1)	Conditional (2)	Unconditional (3)	Conditional (4)
Male	−0.007***	−0.009***	−0.005	0.005
	(0.002)	(0.003)	(0.004)	(0.004)
Age	−0.290***	−0.071	−0.743***	−0.119
	(0.063)	(0.086)	(0.121)	(0.115)
Education	2.070***	0.448***	1.930***	0.007
	(0.056)	(0.088)	(0.090)	(0.118)
Ethnic diversity	−0.253***	0.099***	0.004	−0.010
	(0.026)	(0.033)	(0.044)	(0.027)
Rural	−0.260***	−0.051***	−0.116***	0.008
	(0.009)	(0.017)	(0.016)	(0.022)
Terrain roughness	−0.052*	−0.158**	−0.321***	−0.049
	(0.031)	(0.068)	(0.056)	(0.072)
Log time to nearest cities	−1.299***	−0.327***	−0.324***	−0.008
	(0.026)	(0.036)	(0.052)	(0.054)
Log distance to national borders	−0.060**	0.040*	0.110**	0.029
	(0.025)	(0.022)	(0.048)	(0.021)
Log distance to water bodies	−0.763***	−0.199***	−0.225***	0.021
	(0.026)	(0.026)	(0.050)	(0.029)
Log distance to protected areas	−0.160***	−0.013	−0.153***	0.009
	(0.019)	(0.025)	(0.032)	(0.020)
Log night light density in 2000	2.935***	0.781***	0.798***	0.113
	(0.057)	(0.101)	(0.104)	(0.114)
Log population density in 2000	1.881***	0.383***	0.778***	0.012
	(0.033)	(0.039)	(0.059)	(0.041)

Notes: Columns 1 and 2 compare individual- and cluster-level characteristics between individuals in DHS clusters receiving aid (encompassing both pre- and post-treatment groups) and those in non-aided clusters (the control group). Columns 3 and 4 detail the differences between pre- and post-treatment groups. Unconditional comparisons are in columns 1 and 3, while columns 2 and 4 adjust for country-by-year and ADM2 level fixed effects. Standard errors clustered at the DHS cluster level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

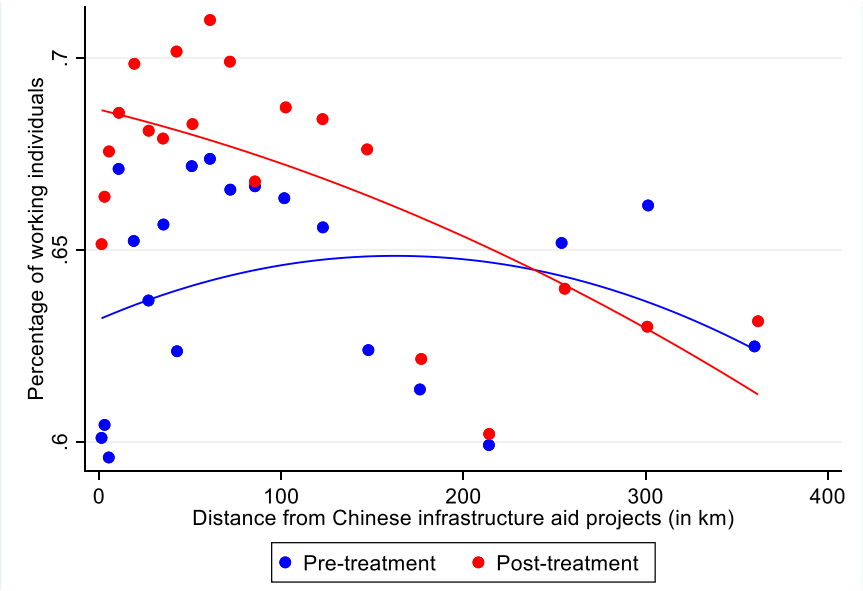


Fig. 2. Employment and distance from Chinese projects.
Notes: The figure categorizes DHS clusters into 20 equal-sized bins, with the local employment average represented by a single point for each bin, separately for post-treatment (red points) and pre-treatment (blue points) groups. The vertical axis measures the percentage of individuals employed within each bin, while the horizontal axis indicates the distance to the nearest Chinese aid project in kilometers. Quadratic fit curves for the post- and pre-treatment data are depicted in red and blue, respectively.

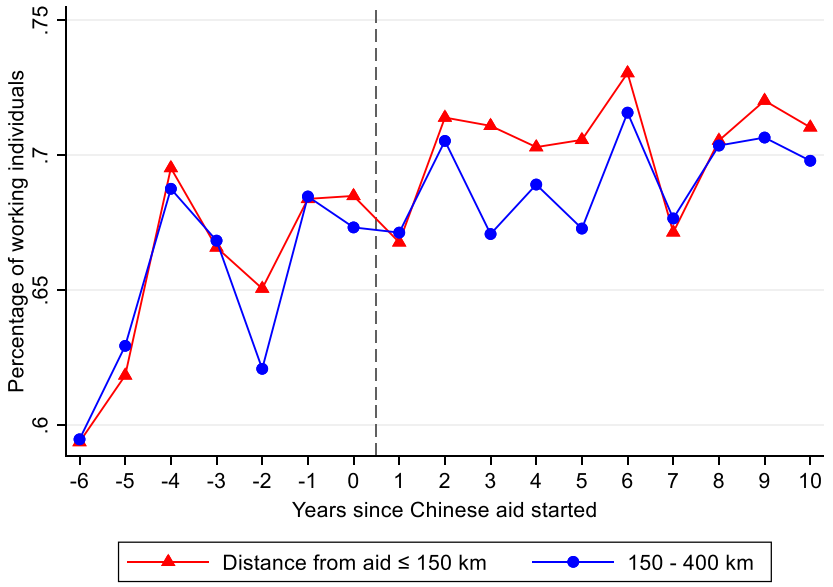


Fig. 3. Employment and timing relative to Chinese projects.
Notes: This figure organizes DHS clusters according to the timings of the DHS survey relative to the initiation of aid projects (horizontal axis) and their proximity to Chinese aid projects. We plot trends for clusters within 150 km of the projects (red line) and for those 150 to 400 km distant (blue line). The vertical axis indicates the employment percentage within each group.

3.1.2. Regression results

Building on the stylized evidence from Figs. 2 and 3, Table 2 formally examines the individual-level employment effects of Chinese infrastructure aid projects. We include ADM2 region and country-by-year fixed effects to control for factors that could influence aid allocation, such as each region's economic and political changes, bilateral relations with China, local ideologies, natural resources, connections with political leaders, and time-invariant regional needs for aid. In columns 2 and 3, we further include individual and cluster-level covariates reflecting the local area's current and potential economic development. Column 3 represents our preferred model corresponding to Eq. (1), incorporating all covariates and fixed effects.

The coefficients for *Post* in the first three columns reveal a two-percentage-point increase in employment in post-treatment areas (within 50 km of implemented Chinese aid projects) compared to control areas (without nearby projects). The *Pre* coefficients suggest that employment in pre-treatment areas does not differ from control areas, implying that there are unlikely project site selections based on existing employment levels. These results align with the patterns shown in Fig. 2, with DiD estimators substantiating at least a two-percentage-point increase in local employment, which represents a 3 % rise from the average employment rate of 0.65, as reported at the bottom of Table 2.

Columns 4–6 include lagged terms for *Post*, with column 6 reflecting our full model of Eq. (2). The *Pre* coefficients show that employment rates prior to project construction are comparable to those in areas without aid projects. However, there is a significant increase of about 1.5 percentage points or a 2 % relative to the mean employment rate in the first year after implementation, escalating to 3 percentage points or 5 % of the mean by the second year, as indicated by the DiD estimators in column 6. This impact diminishes over time. However, considering longer-term effects beyond the fourth year, there is still a noticeable 1.3-percentage-point increase in local employment attributed to Chinese aid projects. Figure B1 confirms these short- and medium-term employment benefits, reinforcing the observations from Fig. 3. While the data show no significant difference in employment between treatment and control groups before the initiation of aid projects, a clear and significant divergence emerges shortly after aid commencement and persists in the medium term.

Appendix Table B1 validates these findings across various empirical specifications. We refine our regression models by including linear time trends for each ADM2 region and fixed effects for each leader-region pair, addressing political patronage influences. Additionally, we utilize a Probit model due to our binary dependent variable and refine our control group selection criteria for more accurate comparisons. Exclusions of specific observations from DR Congo and an analysis limited only to areas near Chinese aid projects further support our baseline results. Moreover, we explore the co-location of Chinese aid projects and their agglomeration effects on local employment in Table B2 of the Appendix. As anticipated, a higher density of infrastructure aid projects in an area

Table 2
Impact of Chinese aid projects on local employment.

Dependent variable	Whether an individual was currently working, mean = 0.65					
	(1)	(2)	(3)	(4)	(5)	(6)
Post	0.018***	0.021***	0.023***			
	(0.007)	(0.007)	(0.007)			
Post_t-4				0.020***	0.021***	0.022***
				(0.005)	(0.005)	(0.005)
Post_t-3				0.010	0.004	0.004
				(0.007)	(0.006)	(0.006)
Post_t-2				0.039***	0.040***	0.041***
				(0.006)	(0.006)	(0.005)
Post_t-1				0.022***	0.024***	0.025***
				(0.007)	(0.007)	(0.007)
Pre	−0.004	−0.000	0.001	0.007	0.009	0.010
	(0.007)	(0.008)	(0.008)	(0.006)	(0.006)	(0.006)
Cluster covariates			Yes			Yes
Individual covariates		Yes	Yes		Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	541,715	536,714	536,699	541,715	536,714	536,699
R-squared	0.084	0.220	0.220	0.084	0.220	0.220
Post – Pre	0.022***	0.021***	0.022***			
	(0.006)	(0.006)	(0.006)			
Post_t-4 – Pre				0.014**	0.011*	0.013**
				(0.006)	(0.006)	(0.006)
Post_t-3 – Pre				0.003	−0.005	−0.005
				(0.009)	(0.009)	(0.009)
Post_t-2 – Pre				0.032***	0.031***	0.031***
				(0.007)	(0.007)	(0.007)
Post_t-1 – Pre				0.016*	0.015*	0.015*
				(0.009)	(0.009)	(0.009)

Notes: Columns 1–3 implement the regression from Eq. (1), while columns 4–6 correspond to the specification of Eq. (2). In both sets, columns 1 and 4 include country-by-year and ADM2 region fixed effects. The individual covariates include gender, age, years of education, and ethnicity dummies. DHS cluster-level covariates include rural/urban status, terrain roughness, travel time to the nearest city, ethnic diversity, distance to capitals, national borders, water bodies, protected areas, natural resource deposits, conflict intensity, population density, and nighttime light density in 2000. Columns 3 and 6 incorporate all covariates and fixed effects from Eqs. (1) and (2). *Pre* refers to individuals within 50 km of a pending project, while *Post* refers to those near a Chinese aid project initiated before the DHS survey. *Post_t-1*, *Post_t-2*, and *Post_t-3* indicate the presence of aid projects within 50 km that started one, two, and three years before the survey, respectively, while *Post_t-4* refers to projects that started more than four years before. DiD estimators are presented at the bottom of the table. Standard errors clustered at the DHS cluster level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

amplifies their positive employment effects.[23]

3.2. Dynamic patterns

In Fig. 4, we examine the temporal impact of Chinese aid using an event-study framework, plotting DiD estimators and 99 % confidence intervals for biennial periods before and after aid project implementation over extended durations.[24] Prior to the start of Chinese aid projects, employment rates in areas within 50 km of a project site do not differ from those in more distant areas, as evidenced by the negligible and statistically insignificant lead term coefficients. This finding upholds the parallel trend assumption inherent in DiD analysis. Post-implementation, local employment surges by three percentage points in the initial two years, after which the effects wane. However, a resurgence in significance is observed from the seventh year onward.

These short-term local employment impacts align with findings at the aggregated levels. Dreher et al. (2021) report that Chinese ODA-like projects significantly bolster economic growth in recipient countries within the first three years after commitment. Similarly, Mueller (2024) notes a 0.34 percentage point reduction in national unemployment rates one year post-project commencement.

[23] We also conducted additional analyses using alternative data and identification strategies, which consistently yield robust findings. For instance, we aggregated the DHS and Chinese aid projects into a grid-cell panel data structure and implemented a standard DiD analysis. Furthermore, we used data from the Living Standards Measurement Study (LSMS) for Ethiopia, Nigeria, Tanzania, and Uganda to measure individual employment. These analyses contribute to demonstrate positive effects of Chinese aid on local employment. The detailed results are available upon request.

[24] We focus on the ten-year periods before and after the initiation of local Chinese aid projects. This timeframe is chosen because 89 % of the DHS surveys were conducted within ten years of these projects' commencement, as shown in Appendix Fig. B2. For projects outside this range, we aggregate the data as described in Eq. (2). For example, "≤−9, −10" indicates DHS surveys conducted more than nine years prior to the start of Chinese aid, while "≥9, 10" refers to surveys conducted over nine years after the project began. In our event-study framework, we set the coefficient to zero for instances where local aid projects began in the same year as the DHS surveys, serving as the reference.

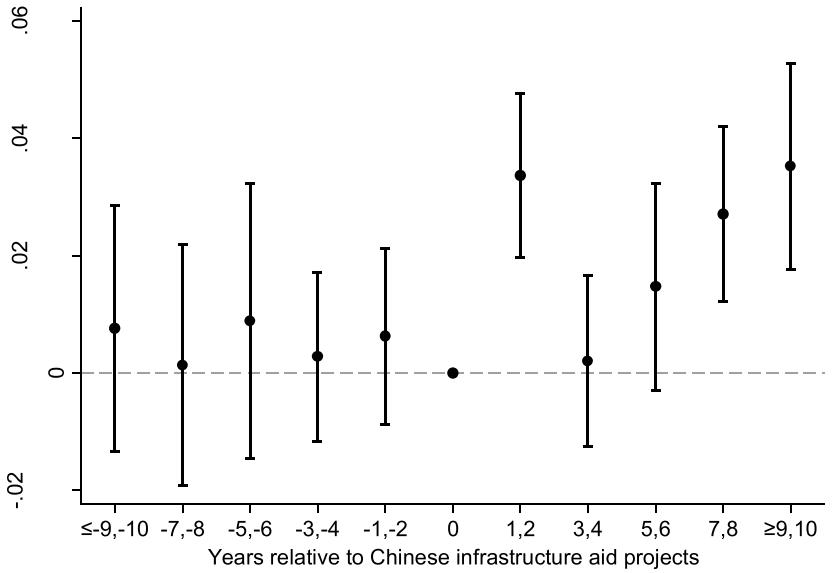


Fig. 4. Dynamic effects of Chinese aid on local employment.
Notes: This figure presents the DiD estimators alongside their 99 % confidence intervals for various two-year intervals before and after the commencement of Chinese aid projects. The two-year period dummy variables are constructed based on the timing of the Chinese aid relative to the DHS, as denoted on the horizontal axis. The regression includes all individual- and cluster-level covariates specified in Eq. (1), as well as the full set of fixed effects. Standard errors are clustered at the level of the DHS clusters.

To dissect the influence of Chinese aid projects on local employment during different stages, we utilize completion data from AidData's Global Chinese Development Finance Dataset, version 2.0 (Custer et al., 2021).[25] From the 255 Chinese aid projects with known completion dates, we define dummy variables *After* and *During* for individuals living within 50 km of a completed project at the time of DHS survey and those near a project under construction, respectively. *Pre* still denotes proximity to a not-yet-implemented project.[26] Thus, the DiD estimator *During* – *Pre* quantifies employment impacts during project construction, while *After* – *Pre* assesses post-completion effects. Table 3 indicates that local employment rises by approximately two percentage points—roughly 3.5 % of the average employment rate—both during and after project construction. This suggests that China-funded large-scale infrastructure projects in Africa not only provide immediate employment opportunities but also foster a sustained employment gains post-completion.

While the short-term effect is likely driven by employment demand created either directly by project construction or indirectly by increased demand from employed workers, the longer-term increase in local employment following infrastructure improvements requires further investigation. Utilizing data from DHS surveys, where respondents indicated their duration of residence at their current location, we analyze migration patterns based on responses from 358,878 individuals. We define three dependent variables to capture whether an individual migrated to the area: (a) within five years, (b) within ten years, or (c) after the aid project's initiation, as shown in columns 1–3 of Table 5. DiD estimations reveal a significant increase in migration: 1.5 percentage points within five years, 2.3 percentage points within ten years, and an even more substantial 2.6 percentage point increase after project commencement.

Column 4 offers an interesting nuance; by excluding recent residents (those who have lived in the area for less than ten years) and focusing on longer-term residents, we find that while the short-term employment impact intensifies, the long-term effect diminishes and lacks statistical significance. This suggests that migrants arriving post-construction are a key driver of the sustained employment impact observed in our baseline results.[27]

These insights complement the broader discussion on the spillover effects of Chinese aid in Section 3.3. Specifically, we note that

[25] In our primary analysis using AidData version 1.1.1, only 115 out of 360 projects (32 %) had available completion dates, with an average project duration of 2.6 years. The subsequent version 2.0 dataset provides more extensive details over a longer timeframe but lacks geocoding for new projects. Therefore, our analysis in Table 3 uses the geocoded projects from version 1.1.1 and supplements completion dates from version 2.0 based on project titles, increasing the number of projects with completion dates to 255 (71 %). We exclude the remaining 105 projects without completion date information to avoid underestimating employment effects, as we cannot accurately assign *Pre, During,* and *Post* statuses to individuals near these projects. To minimize bias, these individuals are omitted from the control group.

[26] To clarify, in defining *Pre, During,* and *Post* variables, we ensure that individuals living within 50 km of a not-yet-implemented project (*Pre*) were not exposed to any projects that commenced prior to the survey. Similarly, for those near an ongoing project (*During*), we confirm no projects within a 50 km radius had been completed before the survey was conducted.

[27] Appendix Table B3 presents descriptive statistics indicating that migrants to areas with Chinese aid projects possess higher levels of education and skills compared to the existing local population.

Table 3
Employment effect of projects at different phases.

Dependent variable	Currently working (mean = 0.65)		
	(1)	(2)	(3)
After	0.016**	0.015*	0.017**
	(0.008)	(0.008)	(0.008)
During	0.017**	0.016**	0.017**
	(0.008)	(0.008)	(0.008)
Pre	−0.007	−0.005	−0.005
	(0.008)	(0.008)	(0.008)
Cluster covariates			Yes
Individual covariates		Yes	Yes
ADM2 FE	Yes	Yes	Yes
Country × Year FE	Yes	Yes	Yes
Observations	524,380	524,019	507,111
R-squared	0.085	0.220	0.221
After – Pre	0.023***	0.020***	0.023***
	(0.07)	(0.007)	(0.007)
During – Pre	0.024***	0.022***	0.023***
	(0.06)	(0.006)	(0.006)

Notes: The variables *Pre, During,* and *Post* are derived from the 255 Chinese aid projects with known completion dates in the AidData's Global Chinese Development Finance Dataset, version 2.0. *Pre* is assigned to individuals within 50 km of a not-yet-implemented project, *During* for those near ongoing constructions, and *After* for individuals close to a completed project at the time of the DHS survey. Number of observations is lower than in baseline results due to the exclusion of individuals near projects lacking completion dates. Individual covariates include gender, age, education, and ethnicity. Cluster-level covariates include rural/urban status, terrain, travel time to the cities, ethnic diversity, distance to capitals, borders, water bodies, protected areas, resource deposits, conflict intensity, population density, and night light density in 2000. Column 3 includes all covariates and fixed effects from Eq. (1). DiD estimators are reported at the bottom of the table. Standard errors clustered at the DHS cluster level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the overall employment benefits of Chinese infrastructure aid are more pronounced near the projects and wane with distance. Additionally, these projects draw migrants post-construction, bolstering long-term local employment without displacing jobs in surrounding areas. This indicates that Chinese aid may activate an underutilized labor force, particularly among the highly educated, in adjacent regions.[28]

Another potential mechanism underlying the long-term effect is the increase in education levels among local individuals, particularly the younger population.[29] This rise in educational attainment equips individuals with the necessary skills and knowledge that enhance their employability, thereby contributing to the significant long-term increase in employment in the affected areas. In Table B4 in the Appendix, we use years of schooling as the dependent variable to estimate how Chinese aid impacts local educational attainment across different age groups. The positive effects on years of schooling are increasingly pronounced for individuals under 30, 25, and 20 years old.

3.3. Spatial spillovers

We analyze the spatial spillover effect of Chinese aid on employment by varying the radius used to define proximity to aid projects, ranging from 25 to 200 km. While our baseline analysis uses a 50 km radius, and we extend this range to better understand the spatial dynamics. The DiD estimators (*Post – Pre*) across these distances are illustrated in Fig. 5, where we observe a stronger employment effect closer to the aid projects, with the most pronounced impact within 25 km. The effect diminishes beyond 100 km, becoming statistically insignificant, suggesting that using a 50 km baseline may underestimate the true employment impact of Chinese aid.[30]

Appendix Table B5 details the dynamic effects based on various radii, showing that while employment impacts decrease with distance, short-term effects remain significant regardless of proximity. This underscores a pronounced spatial spillover during the construction phase, which becomes more localized post-completion.

Focusing on connective infrastructure like roads and railways, which account for a third of the total, we expect and confirm through Appendix Figure B3 a broader spillover effect due to their nature. These projects not only increase employment in adjacent areas but

[28] This is also consistent with the observation that although areas near Chinese aid projects were generally more developed and had more educated individuals (as shown in Table 1), there was no significant difference in employment rates between the pre-treatment and control groups prior to the commencement of aid projects (as indicated by the insignificant coefficients of *Pre* in Table 2).

[29] We are grateful to the anonymous referee for this point.

[30] To protect respondent confidentiality, the DHS geographic displacement procedure allows urban clusters to be displaced by up to two kilometers (0-2 km) and rural clusters by up to five kilometers (0-5 km). The consistent positive employment effect across distances from 25 to 100 km reassures us that our results are unlikely to be affected by these random geographical displacements within local areas. Additionally, we conducted an analysis excluding marginal clusters located between 45 and 50 km from aid projects, and we observe robust results.

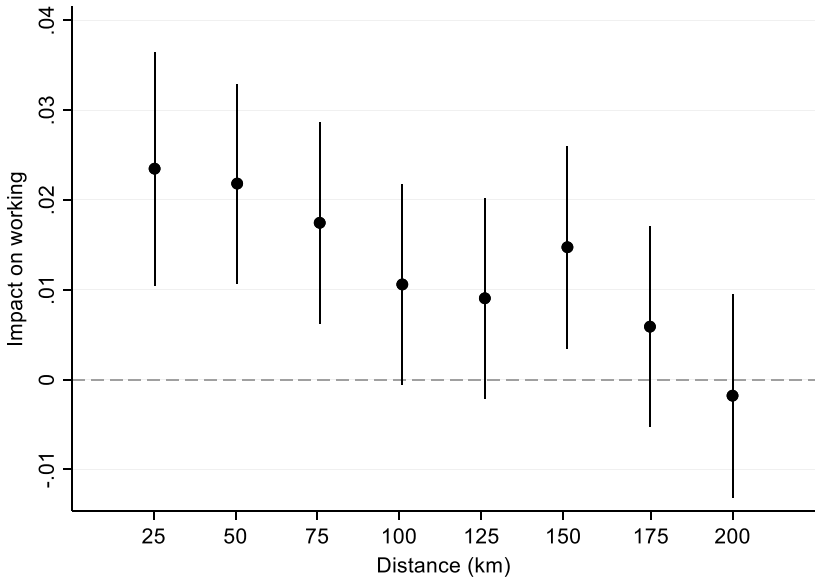


Fig. 5. Spatial spillover of the employment effect.
Notes: On the horizontal axis, the distance of each DHS cluster to the nearest Chinese aid project is plotted, which forms the basis for defining the *Pre* and *Post* variables. The vertical axis indicates the DiD estimator, reflecting the impact on local employment. Each plotted point represents the DiD estimator with its 95 percent confidence interval, derived from a separate regression for distances ranging from 25 to 200 km. These regressions incorporate all individual- and cluster-level covariates as specified in Eq. (1), along with the comprehensive set of fixed effects. Standard errors are clustered at the level of the DHS clusters.

also have a larger average impact, corroborating findings from Bluhm et al. (2024) that Chinese-financed transport projects promote the decentralization in economic activities.

4. Heterogeneous effects

To enhance our understanding of the impact of Chinese aid on employment, we examine the heterogeneous effects based on the type of infrastructure funded by China's official development aid, the educational attainment levels of local individuals, and the varied effects on employment across different occupations, particularly those requiring different labor skills.

4.1. Infrastructure types

We categorize Chinese aid infrastructure projects into four groups: transport, water and power facilities, educational and health, and others. These categories are used to construct *Post*, its lagged terms, and *Pre* variables for our analysis. The DiD estimators presented in Table 4 illustrate the employment impacts for each infrastructure type, with full regression results provided in Appendix Table C1.

For transport projects, encompassing 103 road and railway initiatives, we observe a six-percentage-point increase in employment by the second year, which diminishes to a two-percentage-point increase after four years (as indicated in columns 1 and 2 of Table 4). This declining effect over time corroborates findings of Asher and Novosad (2020), who noted limited medium-term employment benefits within villages resulting from road construction.

Water and power projects, comprising 19.2 % of the initiatives, contribute to a seven-percentage-point rise in employment in the first year, eight points in the second year, and five points after four years (column 4). These results indicate that such facilities not only generate immediate job opportunities but also support sustained employment gains through improved resource access, aligning with literature on the employment effects of electrification in developing countries (Dinkelman, 2011; Lipscomb et al., 2013).

Educational and health infrastructure projects represent 16.7 % of the total and are associated with long-term human capital development. As shown in columns 5 and 6, these projects lead to a three-percentage-point increase in employment in the first year, five points in the second year, and six points after four years, suggesting lasting employment advantages stemming from improved human capital, consistent with the findings of Duflo (2004). The remaining projects (35.6 %), which include communications, government buildings, and other structures like stadiums and demonstration centers, exhibit modest but significant employment effects in

Table 4
Employment effect of different infrastructure.

Dependent variable	Whether an individual was currently working, mean = 0.65							
Infrastructure	Road, railway (28.6 %)		Water, power (19.2 %)		School, hospital (16.7 %)		Public building (35.6 %)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post – Pre	0.031***		0.064***		0.058***		0.009	
	(0.008)		(0.008)		(0.007)		(0.006)	
Post_t-4 – Pre		0.021***		0.046***		0.056***		0.016**
		(0.008)		(0.008)		(0.009)		(0.008)
Post_t-3 – Pre		−0.007		0.056		0.010		−0.002
		(0.013)		(0.043)		(0.013)		(0.013)
Post_t-2 – Pre		0.063***		0.080***		0.051***		−0.006
		(0.011)		(0.011)		(0.008)		(0.008)
Post_t-1 – Pre		−0.006		0.070***		0.030**		0.027*
		(0.015)		(0.013)		(0.013)		(0.015)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	536,699	536,699	536,699	536,699	536,699	536,699	536,699	536,699
R-squared	0.220	0.221	0.220	0.221	0.220	0.221	0.220	0.220

Notes: This table presents DiD estimators from regressions using *Pre, Post*, and the lagged *Post* variables, categorized by different types of Chinese aid infrastructure. Transport infrastructure (roads and railways) is examined in columns 1–2, water and power facilities in columns 3–4, schools and hospitals in columns 5–6, and other public infrastructure types in columns 7–8. *Pre* refers to individuals within 50 km of a pending project, while *Post* refers to those near a Chinese aid project initiated before the DHS survey. *Post_t-1, Post_t-2*, and *Post_t-3* indicate the presence of aid projects within 50 km that started one, two, and three years before the survey, respectively, while *Post_t-4* refers to projects that started more than four years before. Comprehensive regression outcomes are available in Appendix Table B3. Odd-numbered columns apply the Eq. (1) model, while even-numbered columns use the Eq. (2) model. All covariates and fixed effects from these equations are included. Standard errors clustered at the DHS cluster level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5
Migration effect of Chinese aid.

Dependent variable	Whether living in the current residence			Currently working
	for less than 5 years	for less than 10 years	after local aid started	
Sample	Full	Full	Full	Living in the local area for more than 10 years
Mean of Y	0.23	0.34	0.13	0.65
	(1)	(2)	(3)	(4)
Post	0.011	0.015*	0.017**	
	(0.007)	(0.008)	(0.007)	
Post_t-4				0.006
				(0.008)
Post_t-3				0.028***
				(0.010)
Post_t-2				0.041***
				(0.010)
Post_t-1				0.050***
				(0.011)
Pre	−0.004	−0.008	−0.009	0.001
	(0.007)	(0.008)	(0.006)	(0.008)
Covariates	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
Observations	354,615	354,615	354,615	266,938
R-squared	0.122	0.152	0.190	0.252
Post – Pre	0.015**	0.023***	0.026***	
Post_t-4 – Pre				0.005
Post_t-3 – Pre				0.027**
Post_t-2 – Pre				0.040***
Post_t-1 – Pre				0.049***

Notes: The dependent variable in columns 1–3 is binary and denotes migration into the local area within five years, ten years, or after the start of the aid project, respectively. Column 4 focuses on individuals who have resided in the local area for over ten years. *Pre* refers to individuals within 50 km of a pending project, while *Post* refers to those near a Chinese aid project initiated before the DHS survey. *Post_t-1, Post_t-2*, and *Post_t-3* indicate the presence of aid projects within 50 km that started one, two, and three years before the survey, respectively, while *Post_t-4* refers to projects that started more than four years before. The regressions include all individual- and cluster-level covariates specified in Eq. (1), along with a full set of fixed effects. DiD estimators are reported at the bottom of the table. Standard errors clustered at the DHS cluster level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

both the short and long term, as examined in columns 7 and 8.[31]

4.2. Education levels

To explore the impact of Chinese aid on local employment by education level, we categorize individuals based on their highest educational attainment: no primary education, completed primary school, completed secondary school, and education beyond the secondary level. We then examine the interaction between these educational categories and the lagged *Post* terms. The interaction term coefficient, which indicate employment outcome disparities among the educational groups due to Chinese aid projects, are illustrated in Fig. 6, with detailed results in Appendix Table C3.

Our analysis reveals that while the overall employment effects of Chinese aid are significant both during and after project completion, they vary according to educational attainment levels. Individuals with only primary school education or less tend to experience a temporary increase in employment opportunities during the construction phase. In contrast, those with secondary education benefit from more sustained employment prospects in the long term. Notably, individuals with education beyond the secondary level do not see immediate benefits; however, their employment prospects improve significantly in the long term as local infrastructure improves.

4.3. Labor skills

Our investigation into labor skills, as presented in Table 6, reveals significant employment shifts attributable to Chinese aid projects. We assess employment across seven DHS-defined occupational categories, including self-employed and employed individuals in agriculture, sales, service, professional, skilled manual, and unskilled manual labor.[32] The DiD estimates highlight notable changes: agricultural employment decreases by 0.9 percentage points (an 8 % decline from the mean), while service sector employment increases by 0.3 percentage points (an 8 % rise). Additionally, professional, technical, and managerial employment grows by 0.4 percentage points (also an 8 % increase), and skilled manual labor experiences the most substantial growth, rising by 1.7 percentage points (a 24 % increase). These shifts reflect both a potential direct effect on employment driven by construction demands and an indirect boost in sectors such as services.

Interestingly, while unskilled manual labor employment remains stable, skilled labor sees a significant increase following the initiation of Chinese aid projects. Pre-aid project demographics show a predominance of unskilled labor in these areas (as indicated by the *Pre* coefficients in columns 6 and 7), suggesting that Chinese aid may be fostering a shift towards a more skilled workforce.

Further analysis in Table 7 examines the dynamic effects on skilled versus unskilled labor forces. We observe persistent employment growth in skilled labor—both manual labor (column 1) and broader skilled-based jobs (column 3)—following the initiation of Chinese aid.[33] Conversely, unskilled labor experiences a temporary drop in employment (columns 2 and 4), reinforcing the notion of a gradual shift from unskilled to skilled labor over time.[34] Appendix Table C4 corroborates these findings, indicating that formal employment—characterized as year-round jobs and cash wages, often associated with skill development and training—sees the most significant benefits.

Despite criticism that China's overseas investments often favor expatriate workers over local skill development, data from China's official development aid programs paints a different picture. As emphasized in China's Africa policy and foreign aid white papers (State Council, 2015, 2016) and evidenced by the training of over 12 million personnel from developing countries, the increase in skilled labor employment following aid indicates a positive impact in local skill building.

5. Further discussions

In this section, we assess the robustness of our findings in relation to other development resources in the local area and examine the additional impacts of Chinese aid on household wealth and nighttime light density.

5.1. Local developmental resources

While our baseline analyses incorporate a comprehensive set of covariates and fixed effects to control for factors influencing the distribution of Chinese aid, the allocation of such projects may still be confounded by other developmental resources in the area, which are challenging to fully quantify. If the distribution of Chinese aid correlates with other developmental resources (e.g., aid from other

[31] A potential concern is the imbalance in the categories of infrastructure funded by Chinese aid across different recipient countries. For instance, Fig. A2 shows that Nigeria received aid in only two infrastructure categories, while other countries had at least three during the sample period. To address this, we perform a robustness check excluding Nigeria data, as shown in Table C2. The heterogeneous effects remain consistent.

[32] The DHS categorizes occupations according to the major groups defined in the International Standard Classification of Occupations (http://www.ilo.org/public/english/bureau/stat/isco/isco08/).

[33] Following the classification by Hjort and Poulsen (2019), we categorize skilled occupations as those in sales, service, professional work, and skilled manual labor. Unskilled occupations are defined as self-employment in agriculture, domestic work, and unskilled manual labor.

[34] Our results align with existing research on the impact of telecommunications infrastructure, which shows that skilled workers gain more from advancements such as broadband in Norway (Akerman et al., 2015) and fast internet in Africa (Hjort and Poulsen, 2019).

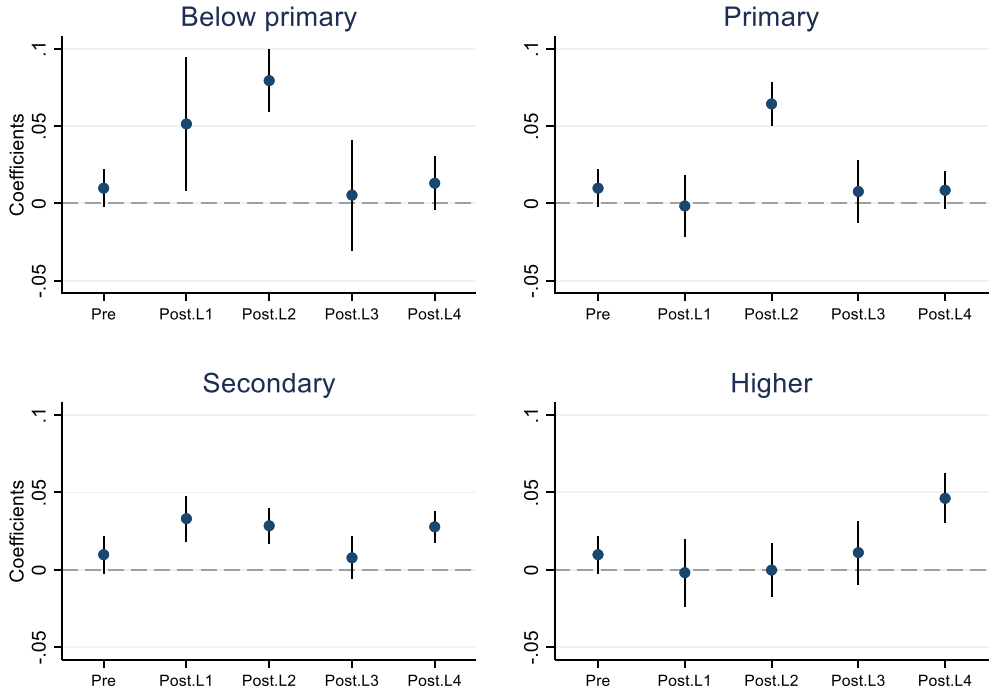


Fig. 6. Impact of Chinese aid projects across education levels.
Notes: Individuals are categorized by educational attainment into four groups, with dummy variables for these groups interacted with lagged *Post* terms. The interaction term coefficients indicate the employment status disparity between individuals in areas without proximate Chinese aid projects and those with varying education levels in clusters within 50 km of active Chinese projects, along with their 95 % confidence intervals, as illustrated in the figure. The regressions account for all individual- and cluster-level covariates as specified in Eq. (1), in addition to the full set of fixed effects. Standard errors are clustered at the level of the DHS clusters.

Table 6
Employment effect for different occupations.

Dependent variable	Agricultural self-employed	Agricultural employee	Sales	Service	Professional work	Skilled manual	Unskilled manual
Mean of Y	0.18	0.11	0.13	0.04	0.05	0.07	0.04
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Post	0.010	−0.009	0.006	0.001	−0.002	0.009***	0.003
	(0.007)	(0.006)	(0.004)	(0.002)	(0.002)	(0.003)	(0.003)
Pre	0.006	−0.000	0.008*	−0.003	−0.006**	−0.007**	0.005*
	(0.008)	(0.007)	(0.005)	(0.002)	(0.002)	(0.003)	(0.003)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	536,699	536,699	536,699	536,699	536,699	536,699	536,699
R-squared	0.400	0.285	0.130	0.060	0.160	0.071	0.095
Post – Pre	0.004	−0.009**	−0.002	0.003*	0.004*	0.017***	−0.002
	(0.006)	(0.004)	(0.003)	(0.002)	(0.002)	(0.003)	(0.003)

Notes: The dependent variable reflects employment status in the occupation specified at the top of each column. *Pre* refers to individuals within 50 km of a pending project, while *Post* refers to those near a Chinese aid project initiated before the DHS survey. The regressions incorporate all covariates and the complete set of fixed effects as specified in Eq. (1). DiD estimators are reported at the bottom of the table. Standard errors clustered at the DHS cluster level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

sources, domestic investment, FDI), their omission from our regressions could bias the estimated effects.

As Chinese aid is primarily demand-driven, recipient country policymakers likely exert considerable influence over the allocation of both Chinese aid and other developmental resources. If these policymakers distribute projects uniformly across regions to promote balanced growth, the employment effects of Chinese aid identified in this study may be underestimated. This occurs because the control group, lacking Chinese aid, might still benefit from other developmental resources that enhance employment. Conversely, if developmental resources are concentrated in select areas to stimulate growth, our estimated impact of Chinese aid could be overestimated if the timing and location of other projects coincide with those of Chinese aid. This latter scenario poses a significant challenge to our main findings, which we address from several perspectives below.

Table 7
Employment effect for skilled and unskilled labor.

Dependent variable	Skilled manual	Unskilled manual	Skilled work	Unskilled work
Mean of Y	0.07	0.04	0.29	0.23
	(1)	(2)	(3)	(4)
Post_t-4	0.008***	−0.004*	0.014***	0.008
	(0.002)	(0.002)	(0.005)	(0.005)
Post_t-3	0.009**	0.006*	0.009	−0.001
	(0.004)	(0.003)	(0.006)	(0.007)
Post_t-2	0.006**	−0.010***	0.017***	0.010
	(0.003)	(0.003)	(0.005)	(0.006)
Post_t-1	0.007**	−0.002	0.029***	−0.018**
	(0.003)	(0.003)	(0.007)	(0.007)
Pre	−0.008***	−0.002	−0.003	0.003
	(0.003)	(0.003)	(0.005)	(0.007)
Covariates	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
Observations	536,699	536,699	536,699	536,699
R-squared	0.071	0.095	0.196	0.375
Post_t-4 – Pre	0.016***	−0.002	0.017***	0.005
	(0.003)	(0.003)	(0.005)	(0.006)
Post_t-3 – Pre	0.016***	0.008*	0.012	−0.004
	(0.005)	(0.004)	(0.008)	(0.010)
Post_t-2 – Pre	0.013***	−0.008**	0.021***	0.007
	(0.003)	(0.003)	(0.006)	(0.007)
Post_t-1 – Pre	0.015***	0.000	0.023***	−0.020**
	(0.004)	(0.004)	(0.008)	(0.008)

Notes: The dependent variables for columns 1 and 2 classify individual as engaged in skilled or unskilled manual labor, respectively. Columns 3 and 4 distinguish between individuals in skilled occupations (including sales, services, professional, and skilled manual labor) and those in unskilled occupations (including self-employed agriculture, domestic, and unskilled manual labor). *Pre* refers to individuals within 50 km of a pending project. *Post_t-1*, *Post_t-2*, and *Post_t-3* indicate the presence of aid projects within 50 km that started one, two, and three years before the survey, respectively, while *Post_t-4* refers to projects that started more than four years before. The regressions include all covariates and the full set of fixed effects as specified in Eq. (2). DiD estimators are reported at the bottom of the table. Standard errors clustered at the DHS cluster level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.1.1. *Other Chinese official flows*

In addition to the ODA-like projects initially analyzed, our sample from ten countries includes 187 Chinese projects classified as OOF-like or VF. Although these projects are less concessional than ODA, they may still contribute to local development, and their distribution might correlate with ODA projects (Dreher et al., 2018). We collected geographic locations and timings for these OOF/VF projects and incorporated *Pre* and *Post* variables for both ODA and OOF/VF projects in our regressions to examine their influence on the baseline effects of ODA projects.

In columns 1 and 2 of Appendix Table D1, we consider only OOF/VF project variables, while columns 3 and 4 include *Pre, Post*, and lagged terms for both OOF/VF and ODA projects. Our findings indicate that OOF/VF projects positively affect local employment, albeit to a lesser extent. When accounting for both project types, ODA projects correspond to a 1.9 percentage point increase in local employment, whereas less concessional OOF/VF projects yield a 1.2 percentage point rise. In terms of short-run effects, OOF/VF projects have a smaller and insignificant increase of 1.4 percentage points in employment in the first year after aid initiation. In contrast, ODA projects significantly boost employment by three percentage points in the second year and by 0.9 percentage points after four years (with marginal significance at p-value = 0.14)

5.1.2. *World Bank aid*

The World Bank is another significant source of development projects. We sourced data on World Bank aid from AidData's World Bank Geocoded Research Release (version 1.4.2), capturing project timings and locations.[35] We include infrastructure projects implemented between 2000 and 2014 in our ten-country sample, identified by project titles. We found 1714 World Bank infrastructure aid projects. Appendix Table D1 shows that World Bank projects positively affect long-term employment but not in the short term, while the short- and long-term effects of Chinese aid persist.[36] In summary, while infrastructure projects funded by less concessional

[35] To our knowledge, there is no project-level data available for other sources of bilateral or multilateral development aid that includes detailed information on both location and timing of implementation. Hence, our analysis here is limited to development aid data from the World Bank.

[36] The observed differences in the temporal patterns between Chinese and World Bank infrastructure projects can be attributed to varying construction timelines. Chinese aid is known for its efficiency in executing large-scale infrastructure projects and demand-driven approach, adhering to the principle of non-interference in recipient countries' internal matters (Dreher et al., 2019). In contrast, World Bank projects typically involve thorough cost-benefit analyses and multiple engagement stages, resulting in longer durations (Dreher et al., 2021). In our dataset, the average duration of World Bank infrastructure projects is approximately seven years, significantly longer than the three-year duration of Chinese projects. This extended timeline may explain the lack of immediate employment effects from World Bank infrastructure aid.

Chinese finance and the World Bank also tend to raise local employment levels, their impacts are considerably smaller and often not statistically significant. In contrast, the effects of Chinese ODA projects remain consistent even when accounting for other projects.

5.1.3. Priority development regions

If the influence of other development resources was mistakenly attributed to the impact of Chinese aid, we would not expect to observe positive effects of Chinese aid in regions not designated as development priorities by governments. Although a precise list of such priority regions in African countries is unavailable, we can infer likely favored areas by policymakers: areas ethnically aligned with political leaders, their birthplaces, and coastal regions. In Appendix Table D2, we exclude these regions and find that the positive impact of Chinese infrastructure aid on employment persist in both the short and long term. Thus, it is unlikely that the employment effects we have observed are solely attributable to other developmental resources.

5.1.4. Proxy variables for local development

Lastly, we collected geocoded data on roads, health facilities, and industrial companies to use their density as proxies for local development levels. In Appendix Table D3, controlling for these densities yields results consistent with our baseline findings.

5.2. Local outcomes

5.2.1. Household wealth

Our analysis extends to the impact of Chinese infrastructure aid on additional local outcomes, specifically household wealth and nighttime light density. Columns 1 and 2 of Table 8 use the household wealth index quintile as the dependent variable, derived from each DHS wave, with values ranging from 1 (lowest) to 5 (highest). Our findings show a 12-percentage-point increase in household wealth (4 % relative to the mean) starting in the third year after the commencement of local Chinese projects, followed by a sustained rise of 7 percentage points (2 % relative to the mean) from the fourth year onward. This trend aligns with our employment findings, suggesting that households accumulate wealth after several years of employment opportunities generated by these projects.

5.2.2. Night light density

Columns 3 and 4 of Table 8 assess the average nighttime light density within a ten-kilometer radius of each DHS cluster during the survey year, with the natural logarithm of this measure serving as the dependent variable.[37] The observations are at the DHS cluster level. The pattern of impact on local nighttime light density mirrors that of employment effects; there is an immediate increase in nighttime light emissions upon the initiation of Chinese aid projects. Specifically, there is a ten-percentage-point rise (11 % relative to the mean) in the first year, followed by increases of three percentage points (4 %) in the second year, and six percentage points (7 %) from the fourth year.[38]

6. Conclusion

This study examines the impact of Chinese infrastructure aid on local employment across ten African countries from 2000 to 2014, utilizing DHS survey data. Our primary approach, akin to a difference-in-differences methodology, compares individuals living near Chinese aid projects with those near pending projects and those in areas without aid. We find that local employment increases by 1.5 to 3.1 percentage points within the first two years of project initiation, followed by a 1.3 percentage point rise in the long term, persisting even after project completion. Notably, while employment effects are evident during and post-construction, they primarily benefit individuals with varying skill sets: less educated individuals in the short term and highly educated workers in the long term. Moreover, completed aid projects attract migration, contributing to local growth.

Our findings support the hypothesis that capital inflows and infrastructure investments spur economic growth in Africa, consistent with classical economic growth theories (Domar, 1946; Romer, 1986; Mankiw et al, 1992; Delong and Summers, 1992). Specifically, while various infrastructure projects temporarily boost local employment, facilities such as schools, hospitals, and water and electricity utilities have a more pronounced and lasting effect on job creation. By utilizing individual-level data and an empirical strategy that leverages temporal and spatial variations in Chinese aid exposure, our study enhances the understanding of the macro-level economic impacts outlined in existing literature.

Furthermore, our methodologically enriches the aid literature by detailing critical steps and assumptions inherent in the difference-in-differences type of approach, particularly relevant for place-based projects in the developing world, where panel data scarcity necessitates such methods. We identify potential pitfalls in the identification strategy and clarify how our approach diverges from a

[37] The data on nighttime light are obtained from NOAA National Centers for Environmental Information, available at https://ngdc.noaa.gov/eog/dmsp/downloadV4composites.html.

[38] The impact on the nighttime light index is notably greater than the effect on employment observed in the baseline results. This finding aligns with Mueller (2024), who analyzed panel data at the recipient country level and suggested that increases in GDP resulting from Chinese aid may not fully translate into improvements in the economic well-being of local citizens, as indicated by the employment rate. First, part of the GDP growth may directly reflect the value of the infrastructure itself. Second, the economic benefits are often disproportionately captured by political elites rather than the general population, largely due to the lack of political conditions attached to aid (Dreher et al., 2019). This alignment underscores the complexity of the relationship between aid and its economic impact.

Table 8
Impacts on other outcomes.

Dependent variable	Wealth index		Night light index	
Mean of Y	3.1		0.9	
	(1)	(2)	(3)	(4)
Post	0.053		0.025	
	(0.032)		(0.019)	
Post_t-4		0.056**		0.061***
		(0.024)		(0.017)
Post_t-3		0.110***		0.028
		(0.026)		(0.035)
Post_t-2		0.015		0.033*
		(0.024)		(0.018)
Post_t-1		0.010		0.098***
		(0.029)		(0.024)
Pre	−0.012	−0.011	−0.026	0.002
	(0.036)	(0.029)	(0.017)	(0.015)
Individual covariates	Yes	Yes		
DHS cluster covariates	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
Observations	554,374	554,374	9592	9592
R-squared	0.616	0.616	0.962	0.962
Post – Pre	0.065**		0.051***	
	(0.025)		(0.014)	
Post_t-4 – Pre		0.068**		0.059***
		(0.028)		(0.019)
Post_t-3 – Pre		0.121***		0.026
		(0.038)		(0.037)
Post_t-2 – Pre		0.021		0.031*
		(0.031)		(0.019)
Post_t-1 – Pre		0.022		0.096***
		(0.040)		(0.025)

Notes: In columns 1 and 2, the dependent variable is the household-level wealth index, categorizing households into wealth distribution quintiles. *Pre* refers to individuals within 50 km of a pending project, while *Post* refers to those near a Chinese aid project initiated before the DHS survey. *Post_t-1, Post_t-2,* and *Post_t-3* indicate the presence of aid projects within 50 km that started one, two, and three years before the survey, respectively, while *Post_t-4* refers to projects that started more than four years before. These columns include all covariates and fixed effects as specified in Eqs. (1) and (2). For columns 3 and 4, the dependent variable is the logarithm of night light density at the DHS cluster level, and the analysis is based on DHS cluster observations, excluding individual-level covariates. Standard errors are clustered at the DHS cluster level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

typical DiD model. A key concern is ensuring that the timing of treatment relative to data collection is genuinely random, which we validate through two verification steps: first, by comparing predetermined characteristics across groups to confirm random selection absent treatment; second, by assessing whether this timing is systematically influenced by local characteristics that change over time. Additionally, we emphasize the importance of incorporating fixed effects correctly to control for confounding factors and ensure accurate group comparisons at appropriate spatial-temporal resolutions. We also address potential reverse causality by comparing groups slated for treatment with those never targeted.

CRediT authorship contribution statement

Jiafu An: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Shiqi Guo:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Haicheng Jiang:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jce.2024.11.003.

References

AidData Research and Evaluation Unit, 2017. Geocoding Methodology, Version 2.0. AidData at William & Mary, Williamsburg, VA. https://www.aiddata.org/publications/geocoding-methodology-version-2-0.
Akerman, Anders, Gaarder, Ingvil, Mogstad, Magne, 2015. The skill complementarity of broadband Internet. Q. J. Econ. 130, 1781–1824.
Alden, Chris, 2005. China in Africa. Survival. (Lond) 47, 147–164.

Anaxagorou, Christiana, Efthyvoulou, Georgios, Sarantides, Vassilis, 2020. Electoral motives and the subnational allocation of foreign aid in Sub-Saharan Africa. Eur. Econ. Rev. 127, 103430.
Asher, Sam, Novosad, Paul, 2020. Rural roads and local economic development. Am. Econ. Rev. 110, 797–823.
Benshaul-Tolonen, Anja, 2024. Industrial gold mining and female empowerment. Econ. Dev. Cult. Change 72, 1213–1266.
Blair, Robert A., Marty, Robert, Roessler, Philip, 2022. Foreign aid and soft power: great power competition in Africa in the early twenty-first century. Br. J. Polit. Sci. 52, 1355–1376.
Bluhm, Richard, Fuchs, Andreas, Strange, Austin, Dreher, Axel, Parks, Bradley, Tierney, Michael J., 2024. Connective financing: chinese infrastructure projects and the diffusion of economic activity in developing countries. J. Urban. Econ. forthcoming.
Brautigam, Deborah, 2011. The Dragon's Gift: The Real Story of China in Africa. Oxford University Press, Oxford and New York.
Brazys, Samuel, Vadlamannati, Krishna C., 2021. Aid curse with Chinese characteristics? Chinese development flows and economic reforms. Public Choice 188, 407–430.
Cervellati, Matteo, Esposito, Elena, Sunde, Uwe, Yuan, Song, 2022. Malaria and Chinese economic activities in Africa. J. Dev. Econ. 154, 102739.
Clemens, Michael A., Radelet, Steven, Bhavnani, Rikhil R., Bazzi, Samuel, 2012. Counting chickens when they hatch: timing and the effects of aid on growth. Econ. J. 122, 590–617.
Cruzatti, C., John, Dreher, Axel, Matzat, Johannes, 2023. Chinese aid and health at the country and local level. World Dev. 167, 106214.
Custer, Samantha, Axel, Dreher, Thai-Binh, Elston, Fuchs, Andreas, Ghose, Siddharta, Lin, Joyce J., Malik, Ammar A., Parks, Bradley C., Russell, Brooke, Solomon, Kyra, Strange, Austin, Tierney, Michael J., Walsh, Katherine, Zaleski, Lincoln, Zhang, Sheng, 2021. Tracking Chinese Development Finance: An Application of AidData's TUFF 2.0 Methodology. AidData at William & Mary, Williamsburg, VA.
DeLong, J.Bradford, Summers, Lawrence H., 1992. Macroeconomic policy and long-run growth. Polic. Long-Run Econ. Growth 77, 93–128.
Depetris-Chauvin, Emilio, Durante, Ruben, Campante, Filipe, 2020. Building nations through shared experiences: evidence from African football. Am. Econ. Rev. 110, 1572–1602.
Dinkelman, Taryn, 2011. The effects of rural electrification on employment: new evidence from South Africa. Am. Econ. Rev. 101, 3078–3108.
Domar, Evsey D., 1946. Capital expansion, rate of growth, and employment. Econometrica 14, 137–147.
Dreher, Axel, Fuchs, Andreas, Hodler, Roland, Parks, Bradley C., Raschky, Paul A., Tierney, Michael J., 2019. African leaders and the geography of China's foreign assistance. J. Dev. Econ. 140, 44–71.
Dreher, Axel, Fuchs, Andreas, Parks, Bradley, Strange, Austin, Tierney, Michael J., 2021. Aid, China, and growth: evidence from a new global development finance dataset. Am. Econ. J.: Econ. Policy 13, 135–174.
Dreher, Axel, Fuchs, Andreas, Parks, Brad, Strange, Austin M., Tierney, Michael J., 2018. Apples and Dragon Fruits: the determinants of aid and other forms of state financing from China to Africa. Int. Stud. Q. 62, 182–194.
Duflo, Esther, 2004. The medium run effects of educational expansion: evidence from a large school construction program in Indonesia. J. Dev. Econ. 74, 163–197.
Fiorini, Matteo, Sanfilippo, Marco, 2022. Roads and jobs in Ethiopia. World Bank Econ. Rev. 36, 999–1020.
Foster, Vivien, Butterfield, William, Chen, Chuan, Pushak, Nataliya, 2009. Building Bridges: China's Growing Role as Infrastructure Financier for Sub-Saharan Africa. World Bank Publications.
Galiani, S., Knack, S., Xu, L.C., Zou, B., 2017. The effect of aid on growth: Evidence from a quasi-experiment. J. Econ. Growth. 22, 1–33.
Gehring, Kai, Kaplan, Lennart C., Wong, Melvin H.L., 2022. China and the World Bank—How contrasting development approaches affect the stability of African states. J. Dev. Econ. 158, 102902.
Go, Delfin S., Page, John, 2008. Africa at a Turning Point? Growth, Aid, and External Shocks. World Bank Publications.
Guillon, Marlène, Mathonnat, Jacky, 2020. What can we learn on Chinese aid allocation motivations from available data? A sectorial analysis of Chinese aid to African countries. China Econ. Rev. 60, 101265.
Guo, Shiqi, An, Jiafu, 2022. Does terrorism make people pessimistic? Evidence from a natural experiment. J. Dev. Econ. 155, 102817.
Hjort, Jonas, Poulsen, Jonas, 2019. The arrival of fast Internet and employment in Africa. Am. Econ. Rev. 109, 1032–1079.
Isaksson, Ann-Sofie, 2020. Chinese aid and local ethnic identification. Int. Organ. 74, 833–852.
Isaksson, Ann-Sofie, Kotsadam, Andreas, 2018a. Chinese aid and local corruption. J. Public Econ. 159, 146–159.
Isaksson, Ann-Sofie, Kotsadam, Andreas, 2018b. Racing to the bottom? Chinese development projects and trade union involvement in Africa. World Dev. 106, 284–298.
Knutsen, Carl H., Kotsadam, Andreas, Olsen, Eivind H., Wig, Tore, 2017. Mining and local corruption in Africa. Am. J. Pol. Sci. 61, 320–334.
Knutsen, Tora, Kotsadam, Andreas, 2020. The political economy of aid allocation: aid and incumbency at the local level in Sub Saharan Africa. World Dev. 127, 104729.
Lipscomb, Molly, Mobarak, A.Mushfiq, Barham, Tania, 2013. Development effects of electrification: evidence from the topographic placement of hydropower plants in Brazil. Am. Econ. J.: Appl. Econ. 5, 200–231.
Mankiw, N.Gregory, Romer, David, Weil, David N., 1992. A contribution to the empirics of economic growth. Q. J. Econ. 107, 407–437.
Martorano, Bruno, Metzger, Laura, Sanfilippo, Marco, 2020. Chinese development assistance and household welfare in Sub-Saharan Africa. World Dev. 129, 104909.
Mueller, Joris, 2024. China's foreign aid: political determinants and economic effects. Working Paper, https://www.jorismueller.com/files/chinaaid_latest_draft.pdf.
Murdock, George P., 1959. Africa: Its Peoples and Their Culture History. McGraw-Hill, New York.
Naim, Moises, 2009. Rogue aid. Foreign. Policy. 159, 96.
Nunn, Nathan, 2008. The long-term effects of Africa's slave trades. Q. J. Econ. 123, 139–176.
Ping, Szu-Ning, Wang, Yi-Ting, Chang, Wen-Yang, 2022. The effects of China's development projects on political accountability. Br. J. Polit. Sci. 52, 65–84.
Perra, Elena, Sanfilippo, Marco, Sundaram, Asha, 2024. Roads, competition, and the informal sector. J. Dev. Econ. 171, 103339.
Qian, N., 2015. Making progress on foreign aid. Annu. Rev. Econ. 7 (1), 277–308.
Ratledge, Nathan, Cadamuro, Gabe, de la Cuesta, Brandon, Stigler, Matthieu, Burke, Marshall, 2022. Using machine learning to assess the livelihood impact of electricity access. Nature 611, 491–495.
Romer, Paul M, 1986. Increasing returns and long-run growth. J. Polit. Econ. 94, 1002–1037.
Sautman, Barry, Yan, Hairong, 2015. Localizing Chinese enterprises in Africa: from myths to policies. HKUST ITEM Thought Leadership Brief Series 2015-05. HKUST Institute for Emerging Market Studies.
Solow, Robert M, 1957. Technical change and the aggregate production function. Rev. Econ. Stat. 39, 312.
Strange, Austin M., O'Donnell, Brian, Gamboa, Daniel, Parks, Bradley, Perla, Charles, 2014. AidData's Methodology for Tracking Under-Reported Financial Flows. AidData, Williamsburg, VA.
Sun, I.Yuan, Jayaram, Kartik, Kassiri, Omid, 2017. Dance of the Lions and Dragons: How Are Africa and China Engaging, and How Will the Partnership Evolve? McKinsey & Company.
The State Council, 2015. China's Africa Policy. The People's Republic of China.
The State Council, 2016. The right to development: China's philosophy, practice and contribution. The People's Republic of China.
Wegenast, Tim, Krauser, Mario, Strüver, Georg, Giesen, Juliane, 2019. At Africa's expense? Disaggregating the employment effects of Chinese mining operations in Sub-Saharan Africa. World Dev. 118, 39–51.
Xu, Zhicheng, Zhang, Yu, Sun, Yang, 2020. Will foreign aid foster economic development? Grid panel data evidence from China's aid to Africa. Emerg. Market. Finance Trade 56, 3383–3404.
Zhao, Suisheng, 2014. A neo-colonialist predator or development partner? China's engagement and rebalance in Africa. J. Contemp. China 23, 1033–1052.